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TOPICAL REVIEW

A Systematic Mapping of Dynamic Thermal Rating for Underground Cables

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ABSTRACT Dynamic Thermal Rating (DTR) offers a flexible and adaptive solution to overcome the limitations of static thermal rating in power transmission. This systematic review aims to map and analyze the current state of research on DTR, specifically focusing on underground cable systems. The review contributes to a clearer understanding of DTR's role in improving the efficiency and resilience of underground cable networks, while also laying the groundwork for future studies to address current limitations. The novelty of this paper lies in its structured synthesis that links modeling techniques, operational improvements, and long-term system reliability. Guided by the PRISMA methodology, 12 relevant articles published between 2005 and 2025 were selected from Scopus and Web of Science databases. These articles were thematically analyzed across three main categories: Population–Intervention–Outcome, Methods–Objectives, and Outcomes–Results. The findings indicate that DTR enhances cable loading capacity, improves temperature prediction accuracy, and supports more efficient network planning and investment strategies. Key methods included thermal modeling, quasi-dynamic simulations, machine learning, and real-time optimization. Several studies reported capacity improvements of up to 22% and cost savings of up to 15%. The review also highlights recent innovations, such as the integration of climate projection models, diagnostic frameworks, and AI-based forecasting techniques.

INDEX TERMS Dynamic rating, systematic mapping, systematic review, thematic, thermal rating, underground cables.

I. INTRODUCTION

In the realm of sustainable energy systems, robust power cables, whether overhead lines or underground cables, stand as critical infrastructure, facilitating dependable and efficient electricity transmission across diverse landscapes [1].

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To optimize the performance of these assets, dynamic cable rating (DCR) and dynamic line rating (DLR) have emerged as pivotal solutions, addressing the limitations of static line rating (SLR) [2] and offering a more flexible and efficient approach to managing the capacity of power cables [3]. While SLR prescribes a constant maximum current capacity under assumed worst-case scenario conditions, DCR and DLR incorporate dynamic changes in environmental

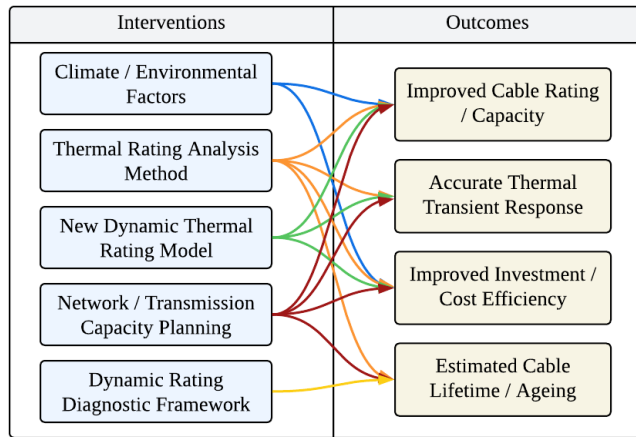


FIGURE 1. Conceptual framework for this study showing the interconnections between interventions and outcomes derived from the analysed articles in dynamic thermal rating studies of underground cables.

conditions and load profiles to better utilize existing cables, often exceeding the standard rating. This enables utility companies to safely enhance power flow in cables under optimal conditions, improving grid efficiency and reliability in line with Sustainable Development Goals (SDG7's) objective of ensuring universal access to affordable, reliable, sustainable, and modern energy [4].

Despite these advantages, existing thermal models for DCR and DLR, which represent cable properties and heat transfer, have notable limitations affecting their accuracy and reliability. These models are highly sensitive to weather [5] and surrounding soil conditions [6]; unaccounted variations in these parameters can lead to inaccuracies in predicting a cable's thermal behavior. Furthermore, accurate thermal models tend to be sophisticated and computationally demanding, potentially limiting their practicality for real-time applications due to factors such as conductor radial temperature [7]. Consequently, while thermal models are useful for estimating ampacity (the current-carrying capacity within safe temperature limits), their predictive reliability is not guaranteed unless they are supported by robust implementation strategies and real-time monitoring data [8].

Given these constraints, how can the research community synthesize the diverse findings on DCR and DLR into a high-quality evidence base that reduces bias, improves reproducibility, and identifies the most effective approaches for underground cable applications? Although numerous studies have explored the benefits and advanced techniques for implementing reliable DCR and DLR systems [9], [10], [11], existing reviews often lack the rigor of systematic literature reviews, which follow structured, transparent methodologies to identify, evaluate, and synthesize all relevant research.

To address this gap, this study conducts a systematic mapping as part of a systematic review on DLR and DCR, which hereafter referred to collectively as dynamic thermal rating (DTR) [9], [10], due to their shared analytical approaches

TABLE 1. Keywords for method of search.

POPULATION (P)	INTERVENTION (I)	OUTCOME (O)
Power cable	Dynamic thermal rating	Ampacity
Underground cable	Dynamic line rating	Current-carrying capacity
Medium voltage cable	Dynamic cable rating	Thermal performance
High voltage cable	Real-time thermal rating	Power delivery efficiency
MV cable	Temperature-dependent rating	Cable electrical capacity
HV cable	Ampacity modelling	Cable loading capacity
Transmission line	Current rating calculation	Thermal rating
Electrical cable	Dynamic current rating	Continuous current rating
Power conductor	Adaptive ampacity	Permissible current
Power line		Thermal current limit
Buried cable		Allowable current

TABLE 2. Inclusion and exclusion criteria.

Criteria	Inclusion	Exclusion
Year	2005 – 2025	<2005
Document Type	Article	Conference Proceeding, Book, Book Chapter, Review
Document Language	English	Chinese, German, Italian, Korean, Polish

and focus on thermal factors influencing current rating. The review evaluates these factors using both objective and subjective methods, providing trustworthy and comprehensive insights [12]. The systematic mapping identifies weaknesses in the field and guides future research by deriving thematic gaps through synthesis and analysis. This process groups data into themes based on the intersections of populations, interventions, and outcomes; methods and objectives; and outcomes and results. The central research question addressed is: How do dynamic thermal rating approaches for underground cables intersect with system populations, technical interventions, and intended outcomes across existing research? As part of this study, Figure 1 illustrates the conceptual framework representing the interconnections between interventions and outcomes derived from the selected articles following the overall methodology.

However, the result from this review is limited on DTR for underground cables and may not capture all cable types, installation methods, or environmental conditions. Most

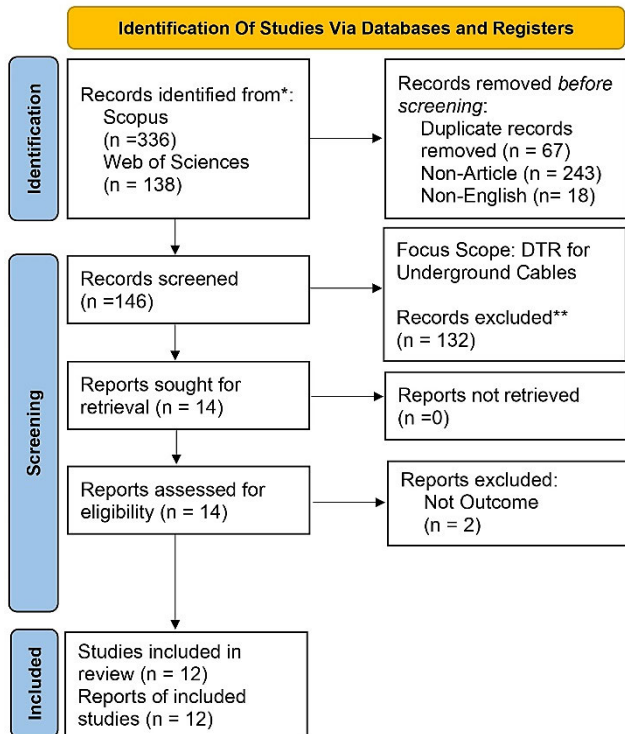


FIGURE 2. Search strategy of the systematic review in this study (PRISMA 2020 flow diagram for new systematic reviews which included searches of databases and registers only).

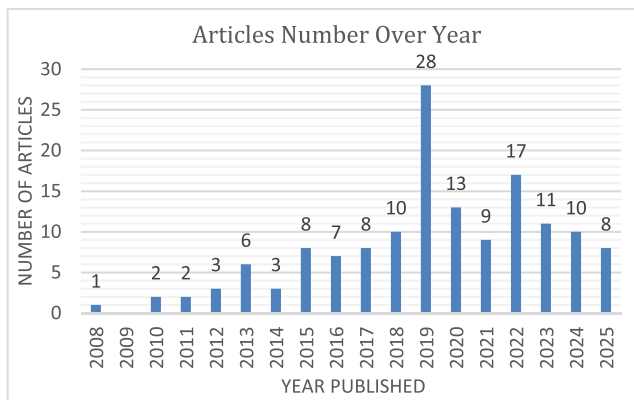


FIGURE 3. Trends in annual publications on dynamic thermal rating.

studies relied on modeling assumptions, which may differ from real-world behavior. Variations in data quality across studies may also affect the results. Despite these limitations, the findings offer practical insights for implementing DTR in underground cable systems and direction for future research.

This paper is organized into several key sections. First, Section I sets the stage by introducing the background, motivation, and goals of the study, emphasizing why DTR is important for underground cable systems. Next, Section II explains how the relevant articles were selected, outlining

the search process, screening criteria, and the PRISMA-based approach used to ensure a structured and transparent review. Section III shares the main results from the thematic analysis, organized into three parts, which are the link between population, intervention, and outcome; how different methods align with research objectives; and how outcomes relate to actual results. Each theme is supported with detailed explanations and summary tables. Furthermore, Section IV highlights the research gaps that emerged from the review and suggests areas for future exploration. Finally, Section V wraps up the paper with a summary of key insights, a reflection on the study's contributions, and a reminder of how the review aligns with its original objectives.

II. MATERIALS AND METHODS

A. METHODOLOGICAL OVERVIEW

The full systematic review is conducted in accordance with the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) 2020 guidelines [13]. PRISMA is a widely recognized standard protocol designed to ensure methodological rigor, transparency, and consistency throughout the systematic review and meta-analysis process. It provides a structured approach that guides researchers through each stage of the review, from identification and screening to eligibility, inclusion, and synthesis of relevant studies [14]. By adhering to the PRISMA framework, this review aims to ensure the completeness and quality of the reporting, minimize the risk of duplication, and reduce potential reporting bias. Ultimately, this approach supports the overarching goal of providing a reliable synthesis of the existing literature in response to the research question.

B. LITERATURE DATABASES

In the preliminary stage, this review seeks to identify published academic articles related to dynamic cable rating (DCR) and dynamic line rating (DLR). For the purposes of this study, the term dynamic thermal rating (DTR) is used as a general descriptor that encompasses both DCR and DLR, given their shared methodological approaches and analytical characteristics. Moreover, the systematic and methodical literature search was conducted using two major electronic databases, which are Web of Science (WoS) and Scopus, due to their extensive coverage of academic literature and their ability to yield highly relevant search results. The search strategy included the conventional keyword queries. The specific keywords used in the search process are presented in Table 1, outlining the detailed search methodology. For this review, the scope is further narrowed to focus specifically on DTR studies related to underground cables to evaluate through either objective or subjective analytical methods.

C. SEARCH TERMS

In this review, PIO (Population-Intervention-Outcome) is used as a basic framework to structure clinical research questions and to guide literature searches as shown in Table 1.

TABLE 3. Summary of recently published articles (2025).

Theme	Ref.	Focus Area	Key Findings
Climate Impact and Cable Behaviour	Sergio M.-S.; Andrea M. [15]	Regional climate impact on transmission systems	Quantifies transmission capacity reductions under RCP 8.5; proposes quasi-dynamic thermal rating; shows seasonal improvements in ampacity up to 22%.
	Sergio, MS; Andrea, M. [16]	Climate modeling, machine learning (ML) for soil parameters	Uses climate models and ML to project soil conditions; up to 9.3% improvement in ratings; shows importance of dynamic adjustment under future climates.
Thermal Modeling Advances	Poradowski W.; Anders G.J. [17]	Finite Element Method (FEM) and analytical modeling of sheath resistance	Develops an analytical model for corrugated sheath layers based on FEM; IEC 60287 is shown to overestimate ampacity; accounts for all heat transfer modes.
	Poradowski W.; Anders G.J. [18]	FEM and analytical modeling of submarine cables	Evaluates thermal performance of submarine cables in bend stiffeners; develops Galerkin-based model; shows convection omission may cause pessimistic ampacity predictions.
High Voltage Systems and Corona Losses	Riba J.-R.; Moreno-Eguilaz M. [19]	Corona discharge and heat modeling in HV lines	Demonstrates that corona losses may significantly reduce ampacity under pollution; recommends inclusion of corona effects in DLR modeling for accuracy.
Experimental and Real-Time Applications	Vejo L. et al. [20]	Real-time monitoring of High Temperature Low Sag (HTLS) conductors	Compares static and DLR results for HTLS conductors; confirms effectiveness of DLR via physical simulations under varying conditions.
	Li J. et al. [21]	Probabilistic overload risk under real-time DLR	Proposes an analytical risk framework using SORM; accounts for uncertainties (weather, failure, load); supports secure real-time dispatch decisions.
Artificial Intelligence (AI) and Forecasting	Yin W. et al. [22]	Artificial Intelligence (AI), forecasting, hydrogen storage integration	Models DTR interaction with storage and extreme weather; uses AI-based stochastic optimization to manage energy imbalance and uncertainty under heat waves.
Thermal-Aging and Cable Diagnostics	Poradowski W.; Anders G.J. [18]	Long-term thermal aging under constrained geometry	(Also fits under diagnostics) Highlights long-term degradation risks in offshore cable sections; supports diagnostics via detailed thermal modeling.

1) SCREENING STRATEGY

The screening process consists of two main phases: (1) title and abstract screening and (2) full-text screening. Both phases are conducted independently by two reviewers. In the first phase, titles and abstracts are screened for relevance based on predefined inclusion and exclusion criteria. Articles that meet the criteria are retained for full-text screening. A pilot screening is conducted prior to the main screening to assess consistency between reviewers. Thirty articles are randomly selected and independently screened by two reviewers. A consistency check using Cohen's Kappa statistic is performed, with a minimum agreement threshold set at 75%.

Following the pilot, the remaining titles and abstracts are screened independently and in parallel by the same two reviewers. When disagreements occur, articles are marked as "Maybe" and referred to a third independent reviewer for

final decision. In the second phase, full texts of all retained articles are retrieved and screened to confirm final inclusion. Abstracts of the full texts are reviewed collaboratively by all reviewers to determine eligibility for data extraction. Any discrepancies are flagged and resolved through group discussion among all six reviewers to reach a consensus.

Full-text screening and data extraction are conducted using a structured Google Form. Articles selected for inclusion are distributed among the six reviewers for data extraction. Articles excluded at this stage are recorded along with the reason for exclusion. All extracted data are cross-checked by one author (KNF) to ensure accuracy and coherence.

2) ELIGIBILITY CRITERIA

To be included at both title and abstract screening and full text screening stages, articles must meet the eligibility criteria described below. These criteria are based on the

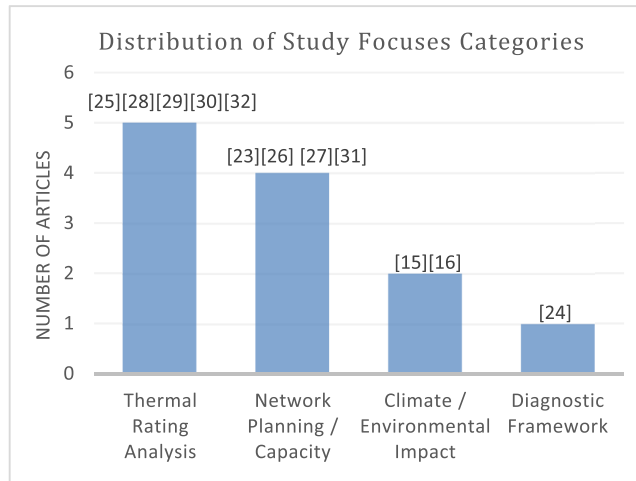


FIGURE 4. Focus of studies for dynamic thermal rating for underground cables articles.

population, intervention, outcome, and study type elements of the systematic map.

a: RELEVANT POPULATION

This systematic map includes studies that focus on cables used in electrical networks, including underground cables, medium voltage (MV) and high voltage (HV) cables, buried cables, transmission lines, and other forms of electrical power conductors. These cables are critical to both urban and rural infrastructure, especially in underground installations where heat dissipation is limited, and operational conditions are more challenging to manage compared to overhead systems. Eligible systems must involve the cable systems. Studies not involving the relevant population are not included unless they clearly describe a setting that incorporates the population elements.

b: RELEVANT INTERVENTION

Eligible interventions must encompass methods that enable more accurate, real-time determination of a cable's current-carrying capacity or dynamic thermal rating. These include dynamic cable rating, real-time thermal rating, temperature-dependent rating, and other forms of dynamic thermal modeling. These approaches adjust the permissible current based on actual environmental and loading conditions, rather than fixed assumptions.

c: RELEVANT COMPARATOR

This systematic map does not require a formal comparator for inclusion. Studies will be considered eligible whether they with or without comparison groups. When present, comparators may include spatial contrasts (e.g., sites with and without interventions), temporal contrasts (e.g., before and after implementation), or system comparisons. Comparative analysis, although not required, will be recorded when available.

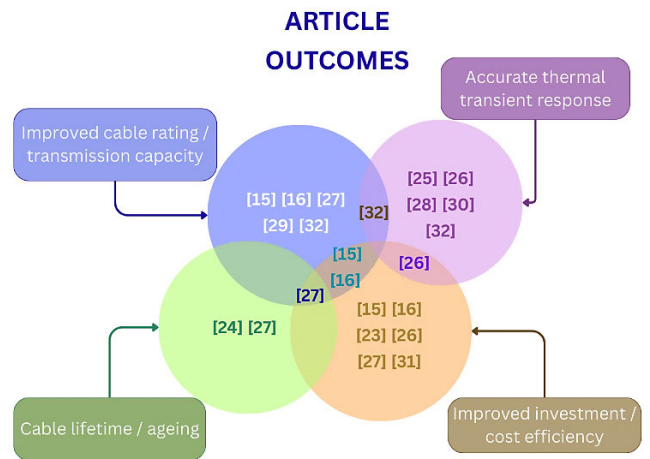


FIGURE 5. Overlap of selected articles according to their reported outcomes.

d: RELEVANT OUTCOMES

Studies must report such interventions include improved ampacity, thermal performance, and overall power delivery efficiency. By dynamically estimating thermal ratings and allowable currents, DCR helps utilities operate closer to the true capacity of their assets, enhancing cable loading capacity while maintaining safety and performance. Outcomes may be reported through quantitative data, observational assessment, or model-based estimation. Studies focusing solely on productivity, economics, or social factors without linking to relevant outcome variables will be excluded unless other metrics are also reported.

e: RELEVANT STUDY TYPE

Empirical studies using observational, experimental, modeling, or quasi experimental approaches will be included. These may be:

- Field studies
- Laboratory studies
- Modeling studies calibrated with real world data
- Case studies with described methodology

Studies must be based on original data. Theoretical frameworks, conceptual discussions, editorials, commentaries, and narrative reviews will be excluded. Only peer reviewed journal articles published in English will be included to ensure reliability and comparability.

3) STUDY VALIDITY ASSESSMENT

The objective of this systematic map is to map and analyze the current state of research on DTR, specifically focusing on underground cable systems, with particular attention to the types of interventions, geographic coverage, and outcome metrics reported. As such, a formal assessment of study validity will not be conducted beyond the eligibility criteria described above.

TABLE 4. Final articles retrieved from systematic searching.

Ref.	Article Title	Authors	Year	Country
[15]	Assessment of the influence of climate change on power grid transmission capacity	Sergio, MS.; Andrea, M.	2025	France
[16]	Long term climate-driven underground cable thermal ratings for network planning	Sergio, MS.; Andrea, M.	2025	France
[23]	Active Distribution Network Expansion Planning with Dynamic Thermal Rating of Underground Cables and Transformers	Li, YX; Chen, QX; Strbac, G; Hur, K; Kang, CQ	2024	China, London, South Korea
[24]	A Diagnostics Framework for Underground Power Cables Lifetime Estimation under Uncertainty	Aizpurua J.I.; Stewart B.G.; McArthur S.D.J.; Jajware N.; Kearns M.; Garro U.; Muxika E.; Mendicute M.	2021	Spain
[25]	Dynamic thermal analysis for underground cables under continuously fluctuant load considering time-varying van wormer coefficient	Wang P.; Kang L.; Xu Z.; Zhou W.; Bao S.; Wu Z.; Hao Y.; Liu G.	2021	China
[26]	Investment Deferral of Feeder Upgrades Revealed by System-Wide Unbalanced Dynamic Rating: Harvesting the Hidden Capacity of Distribution Systems Discovered by Thermal Map Technology	Borbuev A.; Wang W.; Lu H.; Jazebi S.; De Leon F.	2021	United States
[27]	Risk-based underground cable circuit ratings for flexible wind power integration	Liu S.; Kopsidas K.	2021	China, United Kingdom
[28]	Dynamic Thermal Analysis of High-Voltage Power Cable Insulation for Cable Dynamic Thermal Rating	Wang, PY; Ma, H; Liu, G; Han, ZZ; Guo, DM; Xu, T; Kang, LY	2019	China, Australia
[29]	Adaptive soil model for real-time thermal rating of underground power cables	Diaz-Aguiló M.; de León F.	2015	United States
[30]	Introducing Mutual Heating Effects in the Ladder-Type Soil Model for the Dynamic Thermal Rating of Underground Cables	Diaz-Aguilo M.; De Leon F.	2015	United States
[31]	Unlocking distribution network capacity through real-time thermal rating for high penetration of DGs	Degefa M.Z.; Humayun M.; Safdarian A.; Koivisto M.; Millar R.J.; Lehtonen M.	2014	Finland
[32]	Comparison of air-gap thermal models for MV power cables inside unfilled conduit	Degefa M.Z.; Lehtonen M.; Millar R.J.	2012	Finland

However, to enhance the transparency and interpretability of the evidence base, we extract, and code specific attributes related to study design and methodological approach. These include the type of study (e.g., experimental, observational, modeling), the methods used to assess the outcomes, the frequency or duration of performance monitoring, and whether any form of comparator is reported. While not intended as a quality rating, these descriptors support users in making preliminary assessments regarding the strength and applicability of the evidence.

D. SYSTEMATIC SEARCHING STRATEGIES

To ensure the inclusion of relevant studies in this review, a systematic approach was employed, encompassing three primary stages, which are identification, screening, and eligibility assessment. During the identification stage, synonyms, related terms, and variations of the primary keywords relevant to the study were systematically searched. A tailored keyword search string was constructed and applied across selected databases to optimize the retrieval of pertinent literature.

The search process, which was done in April 2025, involved combining keywords (Table 1) through phrase searches and utilizing Boolean operators “OR” and “AND”.

Search String: ((*“power cable*”* OR *“underground cable*”* OR *“medium voltage cable*”* OR *“high voltage cable*”* OR *“MV cable*”* OR *“HV cable*”* OR *“transmission line*”* OR *“electrical cable*”* OR *“power conductor*”* OR *“power line*”* OR *“buried cable*”*))

AND (*“dynamic thermal rating”* OR *“dynamic line rating”* OR *“dynamic cable rating”* OR *“real-time thermal rating”* OR *“temperature-dependent rating”* OR *“ampacity modelling”* OR *“current rating calculation”* OR *“dynamic current rating”* OR *“adaptive ampacity”*)

AND (*“ampacity”* OR *“current-carrying capacity”* OR *“thermal performance”* OR *“power delivery efficiency”* OR *“cable electrical capacity”* OR *“cable loading capacity”* OR *“thermal rating”* OR *“continuous current rating”* OR *“permissible current”* OR *“thermal current limit”* OR *“allowable current”*)).

Figure 2 illustrates the overall process of this study. Starting from databases, two databases (Scopus and WoS) were

used to retrieve the potentially pertinent articles. Every article was checked for duplicates, and those that were found in the databases were eliminated. The publications were screened for titles and abstracts. Included were articles that were deemed pertinent and released within the designated period. The remaining articles were then subjected to a full-text screening. Lastly, exclusion was applied to articles that did not fit the outlined inclusion and exclusion criteria.

1) IDENTIFICATION STAGE

In the search process, publication date filter was also applied, limiting the search to studies published between 2005 and 2025. This approach led to the identification of 474 records from the databases. Next, a total of 474 records were manually screened in accordance with the predefined inclusion and exclusion criteria as shown in Table 2. While literature on DTR in power cables dates back to 1970, this review was limited to studies published between 2005 and 2025, as outlined in the inclusion parameters. At this stage, a total of 213 English-language articles were identified.

2) SCREENING WITH PREDEFINED CRITERIA

The screening process included only English-language technical journal articles, while all non-English (18 papers) and non-technical-article publications (243 papers), such as conference proceedings, books, book chapters, and review papers, were excluded. After removing irrelevant and duplicate entries, 146 records remained. To narrow the focus, only articles specifically addressing DTR for underground cables were considered, resulting in a final selection of 14 relevant studies. These studies were then advanced to the next stage, the eligibility assessment.

3) STUDIES INCLUDED IN QUALITATIVE ANALYSIS

In this stage, particular attention was given to the titles, abstracts, and main bodies of the articles to verify their alignment with the predefined inclusion criteria and the overall objectives of the review. Subsequently, 12 articles that passed this stage underwent further evaluation through qualitative analysis.

E. QUALITY ASSESSMENT

The selected articles were independently reviewed by two subject matter experts using standard quality assessment criteria to evaluate methodological validity and the quality of results. Only peer-reviewed journal articles published between 2012 and 2025 were included, resulting in 12 studies for the final review, comprising 8 high-quality and 4 moderate-quality articles aligns with its original objectives.

Using predefined and consistently applied criteria addressing the scope of analysis, validation approach, system-level relevance, and direct impact on dynamic cable rating outcomes, high-quality studies explicitly assessed system-level implementations such as network planning, operational strategies, and capacity enhancement. In contrast, moderate-quality studies, while scientifically sound, were limited to

component-level modeling or parameter estimation and did not directly evaluate network-wide or planning impacts. Accordingly, the classification reflects differences in contribution scope rather than methodological quality.

III. RESULTS AND DISCUSSIONS

A. OVERVIEW OF ARTICLE PUBLICATION TRENDS OVER THE YEARS

The Figure 3 shows how interest in dynamic thermal rating research has grown steadily from 2008 to 2025. This result shows the distribution of articles for DTR for various population used as in Table 1. While early years saw only a few publications, attention to the topic began to build around 2015 with eight articles. Interest peaked in 2019, with 28 articles published, which is a noticeable spike that likely reflects growing concerns around renewable energy integration, power grid efficiency, and adapting to climate challenges. Since 2019, publication activity has stayed relatively stable, with some ups and downs, but overall reflecting consistent engagement from researchers. From 2020 to 2025, most years saw more than ten studies published, with only slight dips in 2021 and 2024.

Furthermore, the steady flow of work into 2025 highlights how DTR remains a relevant and important area of focus as the energy sector shifts toward more intelligent and flexible infrastructure with eight new studies published. As shown in Table 3, the new published articles, which are published in 2025, explore a wide range of innovations, including improved thermal modeling, real-time rating techniques, and the integration of environmental data to better predict cable temperatures and optimize system capacity. This reflects the growing push toward smarter, more adaptive power systems. A brief summary of the themes and findings from these articles is outlined in Table 3 below.

B. ARTICLES OF DYNAMIC RATING IN UNDERGROUND CABLE STUDIES

Following a systematic search and selection process, a total of 12 finalized articles as shown in Table 4 were selected to be review. The Population was defined to focus only on underground power cables, the Intervention focused on the application of DTR techniques, and the Outcome targeted improvements in thermal performance, operational efficiency, and network reliability. Figure 4 shows the categories of study focuses for the selected articles. Based on these criteria, the selected studies cover a broad spectrum of dynamic rating applications and thermal modeling in underground cable systems.

Articles [15] and [16] by Sergio M.-S. and Andrea M. investigate the influence of climate change and long-term planning using DLR. Articles [23] and [24] explore network planning and cable lifetime estimation under uncertainty. Next, studies [25] and [26] present advanced thermal analysis under fluctuating load conditions and uncover hidden system capacity through thermal mapping.

Article [27] applies risk-based rating to enhance wind power integration, while [28] focuses on thermal performance of high-voltage cable insulation. Articles [29] and [30] introduce soil-based models for real-time thermal rating, addressing adaptive and mutual heating effects. Finally, articles [31] and [32] examine the enhancement of distribution network capacity and the evaluation of air-gap thermal models in unfilled conduit installations. These articles collectively support the review's objective of analyzing the role of DTR in improving the thermal management and performance of underground cable systems.

C. TREND OF STUDY FOR DYNAMIC RATING FOR UNDERGROUND CABLE

Furthermore, Table 4 shows the selected 12 articles span a publication period from 2012 to 2025 and originate from a diverse range of countries, highlighting the global interest in the application of DTR for underground power cables. The most recent contributions from France in 2025 [15], [16] focus on the impact of climate change and long-term thermal planning for underground cables. China, in collaboration with countries such as the UK, South Korea, and Australia, contributes significantly with multiple studies between 2015 and 2024 [23], [25], [27], [28], covering topics from DLR-based expansion planning to high-voltage cable insulation analysis.

A study from Spain [24], in 2021, introduces a diagnostic framework for cable lifetime estimation, while the United States provides critical insights through real-time thermal rating models and system-wide dynamic rating applications [26], [29], [30] in 2015–2021. Finland contributes two earlier but foundational studies [31], [32] in 2012 and 2014 exploring the use of real-time thermal ratings to support distributed generation (DG) and air-gap thermal modeling for cables in conduits. Moreover, to provide a structured understanding of the analyzed articles, a theoretical framework has been developed, as shown in Table 5 below. This framework serves as a guiding model that synthesizes key concepts, methodology, and findings, thereby facilitating clearer interpretation and supporting further research.

D. THEMATIC SYNTHESIS OF POPULATION, INTERVENTION, AND OUTCOMES

The following results sections are based on the thematic synthesis and analysis framework developed for the systematic mapping. From the 12 articles included in this study, the analysis of dynamic rating in underground cable studies reveals a multifaceted approach, addressing various population types, interventions, and outcomes aimed at optimizing power systems.

For Theme 1 of the systematic mapping, the Populations examined include the power grid involving underground cable systems, underground cables, distribution networks, medium-voltage (MV) cables, and high-voltage (HV) cables, as shown in Table 6(a). The Interventions considered are climate or environmental changes, thermal rating analysis, dynamic thermal models, network planning or system

capacity, and diagnostic frameworks. The Outcomes as shown in Figure 6(b) identified consist of improved cable rating or transmission capacity, accurate thermal transient response, enhanced investment or cost efficiency, and improved cable lifetime or aging. As shown in Figure 5, Venn diagram has been created to illustrate the overlap of research outcomes and corresponding articles. Lastly, Table 7 summarize the intersection among Population, Intervention, and Outcome of the articles.

From Table 6(a), the climate and environmental impacts on the power grid were primarily addressed in [15], where Sergio and Andrea evaluated how rising temperatures and climate variability affect power transmission capacity. Their study emphasized the importance of climate-resilient planning for grid infrastructure and the integration of climate-driven data into network planning strategies. Extending this focus to underground systems, [15] and [16] investigated how long-term climate evolution influences underground cable ratings, highlighting the need to adjust cable design and operation in anticipation of future environmental conditions.

Thermal characteristics of underground cables were thoroughly examined in studies such as [23], [25], [26], [27], [29], [30], and [31]. These included thermal rating integration into distribution planning [23], time-varying dynamic thermal analysis [25], and system-wide thermal mapping [26]. Risk-based circuit ratings under operational uncertainty were also explored in [27]. Soil behavior modeling and mutual heating effects, which enhanced DTR accuracy, were tackled in [29] and [30]. Degefa et al. [31] demonstrated real-time rating applications for distributed generation integration. DTR models for underground cables were further developed in [16], [25], [29], and [30], enabling improved adaptability to environmental and load variability. These DTR approaches were also used in network planning in [16], [23], [26], and [27], contributing to cost-efficient infrastructure utilization. Diagnostic efforts were presented in [24] and [27], focusing on cable lifetime estimation and operational risk management.

At the distribution network level, studies [23], [26], and [31] highlighted how thermal awareness and DTR can optimize expansion planning and unlock hidden capacity. Medium voltage (MV) cable thermal performance was studied in [32], where air-gap modeling in conduits revealed critical implications for temperature prediction and operational safety, along with insights into transient thermal behavior. High voltage (HV) cable dynamics were addressed in [28], which presented a model capturing insulation responses under fluctuating loads to ensure rating accuracy and longevity. Together, these studies emphasize the role of thermal modeling and DTR in supporting efficient, resilient, and future-ready cable infrastructure across voltage levels.

From the Outcome part in Table 6(b), numerous studies demonstrated how advanced thermal modeling and DTR techniques improve cable rating accuracy, transient response, and grid efficiency. Climate-adapted ratings by Sergio and Andrea [15], [16] and risk-based circuit ratings by Liu and

Kopsidas [27] support enhanced transmission capacity and integration of renewables. Moreover, adaptive and mutual soil thermal modeling [29], [30], dynamic simulations under fluctuating loads [25], and thermal mapping technologies [26] refined transient responses. On the other hand, modeling of MV and HV cable insulation behaviors [28], [32] improved safety and accuracy.

DTR approaches also enabled cost-efficient network planning by deferring infrastructure upgrades [23], [26], [31], and better utilization of existing assets [27]. Additionally, diagnostic frameworks [24] and operational risk assessments [27] supported cable lifetime estimation and aging management under uncertainty, improving long-term network reliability. Meanwhile, Figure 5 highlights several overlaps among outcomes, where [32] links cable rating or capacity with thermal response, [15] and [16] connect capacity with investment efficiency, and [26] bridges response with investment efficiency. Notably, [27] spans across three outcomes, which are capacity, investment efficiency, and cable lifetime or aging, indicating its broader contribution to multiple research dimensions.

E. METHODS AND OBJECTIVES IN DYNAMIC THERMAL RATING FOR UNDERGROUND CABLES

For the second thematic synthesis, Table 8 summarizes the intersection of Modeling Methods with Population, Intervention, Objectives, and References, while Table 9 presents the detailed examination of methodological approaches and research objectives across the selected studies. Various modeling methods were applied to achieve distinct goals. As shown in the Table 9, several studies compared dynamic methods with conventional static rating practices. Lumped thermal models and thermal-electrical equivalent (TEE) models incorporating the van Wormer coefficient were utilized by Wang et al. [25] and Wang et al. [28], demonstrating more accurate transient and radial temperature behavior compared to static assumptions. Similarly, ladder-type equivalent circuit models that consider mutual heating and air gaps were developed in [30] and [32], offering refined comparisons against traditional methods.

To observe thermal behavior under dynamic and environmental conditions, lumped thermal models [25], [26], [28], and ladder-type equivalent circuits [26], [32] were employed, with Borbuev et al. [26] providing a system-wide thermal map for distribution networks. Advanced soil models, such as ladder-type models combined with Kalman filtering [29], helped reduce uncertainties in transient soil temperature predictions, further improving thermal analysis accuracy. In efforts to improve system and cable lifetime, probabilistic approaches such as Monte Carlo-based Bayesian filtering combined with TEE and thermal-aging models (based on the Arrhenius model) were applied in [24] and [27]. Additionally, Liu and Kopsidas [27] utilized machine learning techniques, specifically ARMA models, to forecast cable conditions and enhance lifetime predictions under operational uncertainty.

Furthermore, network capacity harvesting and optimization were prominent objectives in studies using quasi-dynamic thermal rating (qDTR) models with climatic projections (RCP scenarios) [15], [16]. This also implied in studies that applying machine learning for predictive capacity management [16], [27], as well as real-time thermal rating coupled with optimal power flow (OPF) solutions, as developed by Degefa et al. [31] where it demonstrated the unlocking of hidden operational capacity in underground cables. Lastly, network planning and distributed generation expansion were addressed using optimization algorithms, including the MBD (Model-Based Decomposition) approach to reduce calculation time and cost-based clustering methods in [23]. Studies [16], [26], and [31] also incorporated dynamic thermal insights into planning strategies, ensuring that thermal capacity constraints were effectively integrated into distribution network expansion efforts.

F. EVALUATED OUTCOMES OF DYNAMIC THERMAL RATING APPLICATIONS: ENHANCING CAPACITY, ACCURACY, AND SYSTEM RELIABILITY

Lastly, this theme synthesizes how DTR research translates into tangible results, particularly in terms of cable performance, system efficiency, and grid modernization. As elucidated in Table 10, the intricate correlation between various results and outcomes reported in the articles is illustrated.

Sergio and Andrea [15], [16] demonstrated that climate-aware planning and adaptive DTR implementation significantly benefit system economics and capacity planning. Specifically, [16] reported a 9.3% improvement in cable current rating under climate-informed DTR compared to static ratings, confirming that environmental data integration can unlock additional capacity. Similarly, Li et al. [23] integrated DTR into distribution network expansion planning (DNEP), showing a 14.7% to 15.1% reduction in investment costs, attributed to enhanced load accommodation without overdimensioning infrastructure. Articles [26] and [31] deployed real-time DTR algorithms, achieving better cable utilization and deferring upgrades, which is in [26], this was supported by modular DTR schemes for underground cables with location-specific inputs, while [31] used a dynamic thermal model coupled with OPF, yielding enhanced flexibility and efficiency. Additionally, [32] emphasized the critical role of conduit modeling, where the presence and configuration of air-gaps in conduits directly influenced thermal outcomes, highlighting the importance of physical installation modeling in DTR frameworks.

Another critical insight was the accurate prediction of transient thermal behavior, as cables rarely operate under constant conditions. Articles [25], [26], [28], [30], and [32] addressed this with various modeling methods. Wang et al. [25] applied lumped thermal modeling with transient analysis to evaluate insulation aging due to variable loads, while [28] used FEM-based time-varying simulations to reflect thermal inertia and cable-soil heat exchanges. In [30], a hybrid DTR-load

TABLE 5. Theoretical framework of dynamic rating in underground cable studies.

Title	Year	Authors	Ref. No.	Focus Area	Population	Methodology	Key Findings
Assessment of the influence of climate change on power grid transmission capacity	2025	Sergio M.-S.; Andrea M.	[15]	Impact of climate change on transmission capacity	European power grid (overhead lines, transformers, underground cables)	Quasi-dynamic thermal rating, historical and climate projections	Reductions in capacity (1.53% for overhead lines, 2.1% for transformers, 0.2% for underground cables) by 2070. Quasi-dynamic rating can improve capacity by up to 22% for transformers and 17% for overhead lines.
Long term climate-driven underground cable thermal ratings for network planning	2025	Sergio, MS; Andrea, M	[16]	Long-term effects of climate and soil dynamics on underground cable capacity	Underground cables in France	Climate models, machine learning, cable dynamic models	9.3% improvement in cable rating compared to static ratings. An additional 10% variation between winter and summer highlights the benefit of seasonal dynamic ratings.
Active Distribution Network Expansion Planning with Dynamic Thermal Rating of Underground Cables and Transformers	2024	Li, YX; Chen, QX; Strbac, G; Hur, K; Kang, CQ	[23]	DTR to support Active Distribution Network Expansion Planning (DNEP)	IEEE 33-node and PG&E 69-node systems	DTR-incorporated DNEP model, Modified Benders Decomposition (MBD) algorithm, cost-based clustering	DTR implementation saves 14.7% and 15.1% of investment costs for the two systems respectively. The MBD algorithm significantly accelerates convergence and reduces calculation time.
A Diagnostics Framework for Underground Power Cables Lifetime Estimation under Uncertainty	2021	Aizpurua J.I.; et al.	[24]	Framework for estimating cable lifetime under uncertainty using a probabilistic approach	Underground power cables	Bayesian state-estimation techniques, physics-of-failure (PoF) models	Ambient temperature measurement errors have a greater influence on lifetime estimates than load measurement errors. Higher conductor temperatures amplify the impact of uncertainties on lifetime estimates.
Dynamic thermal analysis for underground cables under continuously fluctuant load considering time-varying van wormer coefficient	2021	Wang P; et al.	[25]	Dynamic thermal behavior of cables under continuous load fluctuation, improving on the IEC method	Underground power cables	Modified lumped thermal models (thermo-electric equivalent method), real cable experiments	Modified methods more accurately reflect a cable's real thermal behavior under continuous load fluctuation. This helps maximize cable utilization while ensuring reliability and can be integrated into existing real-time systems.
Investment Deferral of Feeder Upgrades Revealed by System-Wide Unbalanced Dynamic Rating	2021	Borbuev A.; et al.	[26]	Unbalanced dynamic rating to reveal hidden capacity in distribution systems	IEEE 37-bus distribution network	Transient electro-thermal model for unbalanced three-phase cables	A new methodology for unbalanced ratings can defer capital investments. For a small system like the IEEE 37-bus network, it could save \$5M by avoiding unnecessary upgrades. The approach provides a visual tool to identify hot spots and underutilized energy routes.
Risk-based underground cable circuit ratings for flexible wind power integration	2021	Liu S.; Kopsidas K.	[27]	Risk-based rating methods for integrating stochastic wind power into the grid	Wind farm underground cables in Scotland	Integration of thermal inertia and aging models into flexible rating strategies	A Flexible Cable Rating method increased wind integration by 21% compared to a static rating while maintaining similar aging to a standard dynamic rating. The FCRSeasonal method was identified as the best balance of risk, complexity, and wind integration.
Dynamic Thermal Analysis of High-Voltage Power Cable Insulation for Cable Dynamic Thermal Rating	2019	Wang, PY; et al.	[28]	Optimizing the transient thermal model of high-voltage cable insulation to improve DTR accuracy	High-voltage power cables	Theoretical method combined with Finite-Element Analysis (FEA)	The optimized model provides more accurate dynamic thermal behavior than the IEC model, especially for high-voltage cables and under large loads. This improved model helps increase cable utilization while maintaining reliability.
Adaptive soil model for real-time thermal rating of underground power cables	2015	Diaz-Aguiló M.; de León F.	[29]	Removing uncertainties in soil properties and ambient temperature for more accurate real-time thermal rating (RTTR)	Underground power cables	Adaptive soil model with an extended Kalman filter	The method removes uncertainties in soil parameters and ambient temperature using an extended Kalman filter, achieving a 97% accuracy within a 1.5% error margin. This robust model allows cables to be used to their maximum capabilities within a smart grid context.
Introducing Mutual Heating Effects in the Ladder-Type Soil Model for the Dynamic Thermal Rating of Underground Cables	2015	Diaz-Aguiló M.; De León F.	[30]	Simplifying the modeling of mutual heating effects between cables for DTR calculations	Underground power cables	Ladder-type soil model with current injection	The new model is a simple, easy-to-use alternative to the complex IEC standard, providing accurate results that are compatible with steady-state calculations. This complements existing tools for reliable real-time thermal computations.
Unlocking distribution network capacity through real-time thermal rating for high penetration of DGs	2014	Degefa M.Z.; et al.	[31]	Using Real-Time Thermal Rating (RTTR) to manage distribution networks with high penetration of Distributed Generators (DGs)	Distribution network components (cables, lines, transformers)	RTTR-based active distribution network management framework	The monetary benefits of RTTR increase exponentially with DG penetration. RTTR allows for increased utilization of DGs and better integration potential for new ones without compromising component aging.
Comparison of air-gap thermal models for MV power cables inside unfilled conduit	2012	Degefa M.Z.; Lehtonen M.; Millar R.J.	[32]	Studying heat transfer mechanisms for cables inside conduits to improve DTR models	MV power cables inside unfilled conduits	Laboratory measurements compared with IEC, Electra, and FEA simulations	Models that consider the actual installation situation, such as eccentric vs. concentric alignment, provide more accurate thermal state estimations. The assumption of concentric alignment is overly conservative.

forecasting model used load history to anticipate thermal stress, improving prediction accuracy. Article [32] extended this by evaluating air-gap effects within conduits, providing design-specific insights for medium-voltage cable systems and verifying simulation accuracy through comparison with experimental benchmarks. These studies collectively validate simulation-based DTR approaches as reliable tools for capturing the complexity of in-situ cable temperature dynamics.

Dealing with data uncertainty and reducing conservative overestimations was another focus area. Sergio and Andrea [15], [16] emphasized how traditional planning

often leads to capacity underutilization due to overestimated margins. Liu and Kopsidas [27] addressed this by implementing Bayesian filtering to enhance temperature estimation accuracy under fluctuating ambient and soil conditions, which contributed to reduced overestimation error and increased planning confidence. Similarly, Diaz-Aguiló and de León [29] incorporated Kalman filters into a ladder-type model, dynamically adjusting to real-time soil temperature data. This resulted in more accurate real-time thermal state estimation, crucial for day-ahead operation decisions and contingency management. These contributions show how

TABLE 6. (a) Interconnections of population and intervention. (b) Interconnections of outcome from populations and interventions.

THEME 1 (Population - Intervention)					
Population \ Intervention	Climate / Environmental	Thermal Rating Analysis	Dynamic Thermal Rating Model	Network Planning/Capacity	Diagnostic Framework
Power Grid	[15]			[15]	
Underground Cables	[15], [16]	[23], [25], [26], [27], [29], [30], [31]	[16], [25], [29], [30]	[16], [23], [26], [27]	[24], [27]
Distribution Network		[23], [26], [31]		[23], [26], [31]	
Medium Voltage (MV) Cables		[32]	[32]		
High Voltage (HV) Cables		[28]	[28]		

OUTCOME			Numbers of Articles	
Improved cable rating / transmission capacity			[15], [16], [27], [29], [32]	5
Accurate thermal transient response			[25], [26], [28], [30], [32]	5
Improved investment / cost efficiency			[15], [16], [23], [26], [27], [31]	6
cable lifetime / ageing			[24], [27]	2

TABLE 7. Summary of the intersection among population, intervention, and outcome.

Population	Intervention	Outcome	References
Power Grid	Climate / Environmental	Improved transmission capacity, investment efficiency	[15]
Underground Cables	Climate / Environmental	Improved cable rating	[15], [16]
	Thermal Rating Analysis	Improved cable rating, transient accuracy	[23], [25], [26], [27], [29], [30], [31]
	Dynamic Thermal Rating Model	Improved cable rating, transient accuracy	[16], [25], [29], [30]
	Network Planning / Capacity	Improved investment/cost efficiency	[16], [23], [26], [27]
	Diagnostic Framework	Cable lifetime/aging	[24], [27]
Distribution Network	Thermal Rating Analysis	Improved cable rating, transient accuracy, investment efficiency	[23], [26], [31]
	Network Planning / Capacity	Improved investment/cost efficiency	[23], [26], [31]
MV Cables	Thermal Rating Analysis	Improved cable rating, transient accuracy	[32]
	Dynamic Thermal Rating Model	Improved cable rating, transient accuracy	[32]
HV Cables	Thermal Rating Analysis	Transient accuracy	[28]
	Dynamic Thermal Rating Model	Transient accuracy	[28]

advanced filtering techniques enhance the robustness and dependability of DTR models, even when input data are incomplete or uncertain.

Finally, DTR was shown to play a vital role in smart grid integration and the accommodation of distributed generation (DG). Liu and Kopsidas [27] and Diaz-Aguiló [29] highlighted how real-time thermal models can support operational decisions when integrating renewables and load variability. Article [31] aligned DTR with optimal power flow (OPF) to enable flexible dispatch and active cable rating management, supporting dynamic system reconfiguration. Article [23] reported that grid modernization objectives, including DG hosting and renewable integration, were supported through DTR-enhanced expansion planning. Additionally, [23] noted that convergence performance in optimization algorithms improved with the integration of DTR, suggesting that not only physical system efficiency but also computational planning efficiency benefits from DTR adoption.

Figure 6 below shows the number of articles for respective results categories. Based on the figure and thematic synthesis from Table 10, the dominant focus of DTR research lies in improving transmission capacity and cost efficiency, with multiple studies [15], [16], [23], [26], [31] demonstrating how DTR contributes to deferred upgrades and optimized investment planning. These outcomes are frequently supported by performance evaluations through simulation or experimental testing [15], [16], [23], [25], [26], [28], [31], [32], as well as detailed thermal behaviour modelling using FEM or FEA or CFD techniques [25], [26], [28], [30], [32]. Such models capture transient thermal responses, enabling more precise cable utilization and validating the effectiveness of DTR strategies under real operating conditions.

However, several research gaps emerge. Notably, cable lifetime and aging analysis remains underexplored, with only limited contributions [24], [27] examining how DTR influences insulation degradation over time. Additionally, computational improvements, such as enhanced convergence rates in optimization algorithms, are only briefly mentioned [23], suggesting a need for further research into algorithmic efficiency. Finally, handling data uncertainty is another underrepresented area; although studies like

TABLE 8. Summary of modelling methods with population, intervention, objectives, and references.

Modeling Method	Population	Intervention	Objective / Outcome	References
Lumped thermal / TEE model (with van Wormer coefficient)	Underground Cables	Thermal Rating Analysis	Comparison with static rating; transient/radial temperature analysis	[25], [26], [28]
Quasi-dynamic thermal rating (qDTR) with climatic projections (RCP model)	Power Grid, Underground Cables	Climate / Environmental	Improved cable rating; capacity harvesting under climate variation	[15], [16]
Optimization (MBD-reduced cost + clustering)	Distribution Network	Network Planning / Capacity	Network expansion planning; reduced computation time and cost	[23]
TEE with thermal-aging (Arrhenius), Monte Carlo Bayesian filtering	Underground Cables	Diagnostic Framework	Cable aging and lifetime estimation under uncertainty	[24], [27]
Machine learning (ARMA probabilistic forecast)	Underground Cables	Thermal Rating and Forecasting	Improve lifetime and rating accuracy under operational variability	[16], [27]
Ladder-type soil model with Kalman Filter	Underground Cables	Thermal Rating Analysis	Reduced uncertainty in soil temp variation; accurate thermal response prediction	[29]
Ladder-type equivalent circuit (mutual heating and air gap)	Underground and MV Cables	Thermal Rating Analysis	Enhanced static method comparison; transient radial behavior observed	[26], [30], [32]
Real-time thermal rating with Optimal Power Flow (OPF)	Distribution Network	Dynamic Thermal Rating and Network Planning	Unlock hidden capacity; improve operational planning and system flexibility	[31]

TABLE 9. Interconnections of methods and objectives of selected studies.**THEME 2 (Methods - Objectives)**

METHODS \ OBJECTIVES	comparison with static rating / conventional method	observe thermal behavior / transient / radial temperature distribution in cable and soil / air	improve system / cable lifetime	harvesting network / cable capacity	network planning / distribution network expansion planning (DNEP) / DGs
lumped thermal / TEE model (with van Wormer coefficient)	[25], [28]	[25], [26], [28]		[26]	
quasi-dynamic thermal rating model (qDTR) with climatic projections / model (RCP)	[16]			[15], [16]	[16]
DNEP model with optimization algorithm (MBD - reduce calculation time) and cost-based clustering method				[23]	[23]
TEE with thermal-aging model (Arrhenius model) / probability density function (pdf) - Monte Carlo based Bayesian filtering method	[27]		[24], [27]		[27]
machine learning / ARMA (probabilistic / forecast value)	[16], [27]		[27]		[16], [27]
ladder-type soil model with Kalman Filter (remove the uncertainties)		[29]		[29]	
Ladder-type equivalent circuit (with mutual heating / air gap)	[30], [32]	[26], [32]		[26]	
real-time thermal rating with OPF	[31]			[31]	[31]

[27] and [29] introduced Bayesian and Kalman filtering, broader application of uncertainty quantification methods

could significantly strengthen the reliability of DTR systems. Addressing these gaps will be essential for supporting

TABLE 10. Interconnections of outcome and results of selected studies.

THEME 3 (Outcomes - Results)						
OUTCOMES \ RESULTS	increase transmission capacity and lower system cost	FEM/FEA/CFD results reflect the cable’s real thermal behaviour	Performance evaluation through experimental / simulation on IEEE-test system / PG&E / OPF	reduces the overestimation errors due to data uncertainty	improve integration of RE/DNEP/smart grid/DGs	improves the convergence rate when solving equation
Cable/components ratings	[15], [16]	[32]	[15], [16]	[16], [27], [29]	[27], [29]	
temperature/thermal transient/radial response		[25], [26], [28], [30], [32]	[25], [26], [28], [32]		[28]	
cost efficiency/investment	[15], [16], [23], [26], [31]		[15], [16], [23], [26], [31]		[15], [23], [31]	[23]
cable lifetime/ageing			[27]	[24]	[27]	

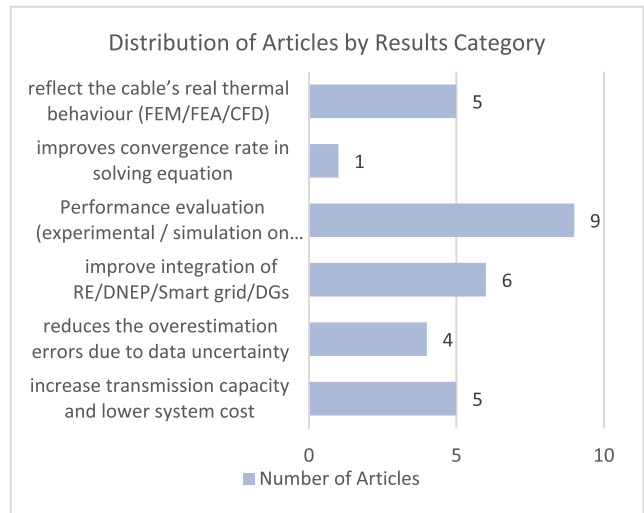


FIGURE 6. Distribution of articles by results category.

more resilient, data-driven, and long-term implementations of DTR in modern power systems. A brief explanation of the research gaps and future research directions is discussed in the subsection below.

IV. RESEARCH GAPS AND FUTURE RESEARCH DIRECTIONS

The systematic mapping of the 12 selected articles revealed that although dynamic thermal rating (DTR) for underground cables has attracted increasing research interest over the past decade, several significant gaps remain.

A. LIMITED INTEGRATION OF REAL-TIME ENVIRONMENTAL VARIABILITY IN DTR MODELS

Firstly, there is limited integration of real-time environmental variability. For example, soil moisture, rainfall patterns, and temperature gradients into existing DTR models. While several studies [16], [25], [29], [30] introduced adaptive soil models and mutual heating effects, only a few incorporated real-time data streams or probabilistic forecasting using

machine learning or filtering techniques like Kalman and Bayesian methods. This lack of operational realism reduces the predictive robustness of DTR systems under dynamic environmental conditions. In the context, which aims to map DTR methods with respect to environmental responsiveness, system performance, and planning reliability, this shortfall highlights the urgent need for more responsive and data-driven DTR frameworks in future.

B. UNDERREPRESENTATION OF MV DISTRIBUTION AND TUNNEL ENVIRONMENTS

Secondly, the review identified an underrepresentation of studies focusing on medium-voltage (MV) distribution networks and cable systems installed in urban tunnel environments. The majority of the reviewed articles [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31] primarily focus on high-voltage (HV) or general underground cable systems, with only one study [32] addressing MV cables in conduit installations. Additionally, very few works consider the unique thermal challenges posed by urban tunnel environments, such as constrained airflow and structural heat accumulation. Given the increasing adoption of tunnel-based cable installations in densely populated urban settings, this omission reveals a critical research gap. Addressing it would support the development of DTR solutions better aligned with real-world operating environments and infrastructure trends, which aligns with the broader objective to promote sustainable and context-sensitive underground cable rating approaches.

C. DISCONNECT BETWEEN DTR AND ASSET MANAGEMENT OBJECTIVES

Lastly, the mapping revealed a disconnect between DTR methodologies and long-term asset management objectives such as insulation aging, degradation modeling, and preventive maintenance planning. Only two studies [24], [27] linked DTR approaches with cable lifetime estimation under operational uncertainty, despite the well-established correlation between thermal stress and insulation deterioration. This

suggests a fragmented research landscape, where thermal modeling and reliability engineering remain largely siloed. Bridging this gap is essential for enabling predictive maintenance, improving asset longevity, and optimizing network investment. As the role of DTR in supporting resilient and future-ready underground cable systems emphasized, the need for integrative frameworks that align real-time thermal insights with long-term infrastructure management is both timely and critical.

V. CONCLUSION

In this study, a systematic review was conducted to synthesize current research on dynamic thermal rating (DTR) in underground cable systems, with a focus on its impact across modeling approaches, technical interventions, and system-level outcomes. Specifically, the review highlights three core research themes, which are the interconnection of population, intervention, and outcomes; methodologies and research objectives; and the outcomes and reported results. The 12 selected studies demonstrated that DTR can increase underground cable transmission capacity by up to 22% and reduce system investment or upgrade costs by approximately 14.7% to 15.1%. These benefits were achieved through a variety of modeling strategies, including lumped thermal models, quasi-dynamic thermal rating based on climate projections, equivalent circuit methods, and machine learning techniques. Several studies also employed real-time DTR integrated with optimal power flow (OPF) models to unlock hidden network capacity and optimize load distribution. Additionally, advanced methodologies such as Kalman filtering, Bayesian probability analysis, and ARMA-based forecasting contributed to improved lifetime estimation, thermal accuracy, and system diagnostics.

Recent innovations highlighted in these studies include the use of AI-driven forecasting, climate-aware DTR using RCP scenarios, and diagnostic frameworks for predicting cable degradation. These advances mark a shift from purely thermal modeling toward more intelligent, predictive, and environment-responsive systems. However, gaps still exist. Real-time environmental data, such as soil moisture and rainfall, are rarely integrated, and MV cables and tunnel installations remain underexplored. There is also a disconnect between short-term DTR modeling and long-term asset management strategies like insulation aging and preventive maintenance. Bridging these gaps requires future work that integrates environmental sensing, dynamic planning, and condition-based diagnostics. Overall, this review successfully meets its objective by systematically connecting modeling methods, system populations, interventions, and outcomes, providing a roadmap for future research and real-world deployment of DTR in modern underground cable networks.

AUTHORS CONTRIBUTION

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Khairunnisa Norfaeza M. K. Fariz and Mohd Noor Afiq Bin Ramlee; Writing—Original Draft Preparation: Khairunnisa Norfaeza M. K. Fariz; Writing—Review and Editing: Mohammad A. Hamdan and Chin-Leong Wooi; Supervision and Critically Revised the Work: Nik Hakimi Nik Ali and Mohd Azraei Pangah @ Pa'at. All authors read and approved the final manuscript.

DATA AVAILABILITY

The authors confirm that the data supporting the findings of this study are available within the article.

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